Independent, parallelizable agents:

1. **Classical MPS/DMRG agent** — ground-state prep via DMRG, bond-dimension convergence tests (χ=16–256), reference DSF/RGF for all four models (XX, $KCuF_3$, two-soliton, $CsCoX_3$)
2. **Ground-state ansatz agent** — variational Hamiltonian ansatz optimization, fidelity vs. system size/layers sweep (Fig S2), singlet vs. Néel initialization branches
3. **AQC compilation agent** — tensor-network approximate compiling per model (each model's AQC run is independent of the others — 4 parallel sub-jobs)
4. **Hardware calibration/layout agent** — mapomatic layout search, layer-fidelity RB experiments, device characterization (can run on ibm_kingston/boston/pittsburgh simultaneously)
5. **Noise-modeling agent** — depolarization-channel MPS simulations across the error-rate sweep (7.5e-3 → 1e-3), independent of real hardware runs
6. **Metrics/analysis agent** — MSE, Wasserstein, SSIM, nQFI, two-tangle code, can be built and unit-tested against dummy spectra before real data exists

Dependent/sequential stages (with outputs from above):

7. **Circuit assembly agent** — merges ground-state circuit (2) + AQC block (3) + Trotter tail → needs 2 and 3 done first
8. **Quantum hardware execution agent** — needs assembled circuits (7) + chosen layout (4); runs PT/DD/TREX pipeline
9. **DSF reconstruction agent** — Fourier-transforms RGF from hardware (8) and from noisy-MPS (5) — can process both once each finishes
10. **Benchmarking/comparison agent** — feeds hardware DSF (9), MPS DSF (1), and experimental INS data into metrics code (6)